

Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI

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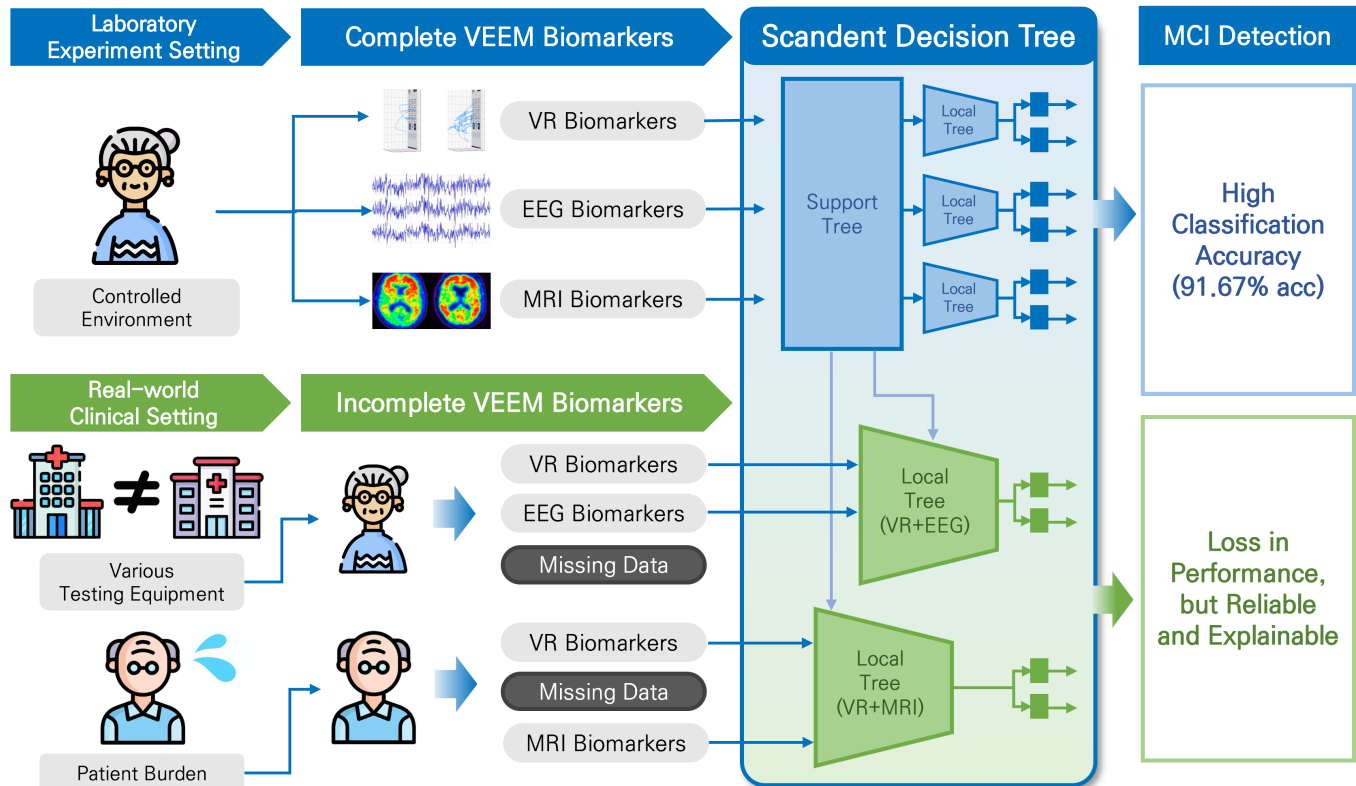


Figure 1: The Overview of This Study

Abstract

Detecting mild cognitive impairment (MCI), an early stage of neurodegenerative disorders, is crucial for timely intervention. This study investigates the potential of a multimodal approach integrating VR, EEG, and MRI (VEEM) biomarkers to enhance MCI detection performance. However, as acquiring all the modalities is challenging, we evaluated the detection performance across complete VEEM biomarkers and datasets with a missing modality. The model using

complete VEEM biomarkers from 59 participants achieved 91.67% accuracy, 85.70% sensitivity, 100.0% specificity, 93.06% precision, and 91.72% F1-score. Notably, the model utilizing VR and EEG with missing MRI biomarkers showed better performance compared to models lacking VR or EEG biomarkers. Our findings suggest a sequential diagnostic approach: VR and EEG biomarkers for early MCI detection, and complete VEEM biomarkers for mid-to-late stage MCI diagnosis. This study underscores the clinical potential of integrating VEEM biomarkers, addressing key challenges related to missing data.

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CCS Concepts

• Human-centered computing → Empirical studies in HCI;
• Applied computing → Health informatics; • Computing methodologies → Machine learning approaches.

Keywords

Mild cognitive impairment, Early detection, Multimodal approach, VR, EEG, MRI, Missing data

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1 INTRODUCTION

Alzheimer's disease (AD) is a neurodegenerative disorder that accounts for 60% to 80% of the total dementia population [16]. As AD is the sixth leading cause of death in the United States, its prevalence continues to rise in tandem with population aging [6]. One promising solution to mitigate the associated societal burdens is the early detection of mild cognitive impairment (MCI), a preclinical stage of AD [12], thereby enabling timely pharmacological or non-pharmacological interventions.

Patients with MCI exhibit multifaceted impairment including cognitive deficits [12], behavioral alterations [24], neurophysiological alterations [22], and neuroanatomical atrophy [30], each of which has a distinct onset range and characteristics. Behavioral data, such as hand and eye movement data obtained through virtual reality (VR), can rapidly and noninvasively differentiate MCI from normal aging [23, 24]. However, a combined analysis of VR and other modalities is essential, as the clinical validity of VR is not yet firmly established [25]. In the case of neurophysiological alterations, electroencephalography (EEG) and evoked potential (EP) can capture neural anomalies in MCI with high temporal resolution, but these techniques are limited in assessing deeper brain structures (e.g., the hippocampus) [31, 32]. Furthermore, EP has been found to be inefficient feature for MCI detection [23]. For neuroanatomical atrophy, magnetic resonance imaging (MRI) is one of the most commonly used methods to examine structural abnormalities by quantifying the matter of subregions in the brain [9]. On the other hand, the prolonged scanning times of MRI, which last about 30 minutes to 2 hours, make it difficult for patients with MCI to tolerate the entire process; moreover, individual differences, such as gender and brain size can confound the distinction between pathological and normal structural changes [5]. Thus, combining these modalities (i.e., VR, EEG, and MRI) through a multimodal strategy could complement the limitations of each modality by leveraging rapid, noninvasive measures (i.e., VR), reliable neural assessments (i.e., EEG), and precise structural insights (i.e., MRI). Consequently, a multimodal approach using VR-EEG-MRI (VEEM) biomarkers shows potential for enhancing the accuracy and efficiency of MCI detection [23].

With the advancement of artificial intelligence, numerous studies have investigated the potential of multimodal approaches in machine learning models. For instance, Park and colleagues [30] demonstrated that machine learning models with a multimodal approach have additional advantages in accuracy and specificity in detecting MCI compared to those utilizing a unimodal approach. However, obtaining complete multimodal data from every patient is often unfeasible in real-world clinical settings due to constraints

such as cost, time, and patient burden [11, 14, 26]. Additionally, data collection procedures and instrumentation can vary across hospitals, leading to inconsistencies in multimodal biomarker datasets. These challenges often result in missing data for one or more modalities, despite machine learning models being trained and evaluated on complete datasets. To address these challenges, recent studies have proposed machine learning techniques to manage missing data in multimodal datasets. For instance, Adarash and colleagues [1] achieved 98.27% accuracy in the classification of AD and MCI using unimodal MRI data with randomly missing values, by employing a scandent decision tree—a method specifically designed to handle missing data [15]. To the best of our knowledge, although the integration of VR, EEG, and MRI biomarkers has been highlighted as essential in real-world clinical practice, the integration of all VEEM biomarkers for MCI detection, particularly in the context of missing data, remains underexplored.

In light of these challenges and opportunities, this study pursues two key objectives. First, to explore the potential of VEEM biomarkers through machine learning models for MCI detection. Second, to provide valuable insights for clinical applications by addressing the issue of missing data within VEEM biomarkers. This research focuses on exploring advanced machine learning techniques for managing incomplete data and integrating multiple biomarkers, with the ultimate goal of developing a more practical and reliable framework for MCI detection to enhance diagnostic accuracy and support timely clinical interventions.

2 RELATED WORK

2.1 MCI Detection Using Multimodal Data

Multifaceted impairment is a hallmark of MCI, encompassing compromised cognitive [4, 33], behavioral alterations [21, 28], neurophysiological deficits [20, 27], and neurostructural atrophy [13, 14]. These alterations may occur concurrently or progress in stages, typically beginning with behavioral manifestations, followed by neurophysiological changes, and ultimately structural anomalies [8, 10]. While prior studies employed unimodal approaches for detecting MCI, focusing on individual modalities such as VR [25, 28], EEG [20, 31], or MRI [14, 19], growing evidence suggests that integrating multiple modalities can substantially enhance the reliability of MCI detection by leveraging complementary information across different datasets [18, 19, 23]. For instance, Kim and colleagues [22] have demonstrated that the machine learning model combining both VR and EEG biomarkers achieved 98.38% accuracy in detecting MCI. Furthermore, Park and colleagues [30] reported improvements when merging VR and MRI biomarkers compared to a unimodal approach. However, few studies have adopted a multimodal approach that integrates behavioral, neurophysiological, and neurostructural data for detecting MCI. Exploiting VR for rapid behavioral assessment, EEG for detailed neural activity analysis, and MRI for robust structural imaging could potentially offer a comprehensive solution for more precise MCI detection. Nonetheless, the complexity of acquiring these diverse data types remains a significant challenge in clinical settings.

2.2 Addressing Missing Data in Multimodal Dataset

In clinical settings, obtaining multimodal data is often challenging due to various constraints such as high costs, limited equipment availability, and patient burden, often leading to incomplete data acquisition. To address these limitations, various methodologies have been proposed to handle missing data in multimodal datasets [2, 7]. For example, Hor and Moradi [15] introduced a scandent decision tree, which yields reliable diagnostic outcomes even in the presence of missing data. This method incorporates a support tree trained on complete datasets and local trees trained on incomplete datasets with missing data. By aligning the local tree training with the patterns established by the support tree, this methodology not only reduces performance loss but also enhances the interpretability of local tree outputs within the unified context of the support tree. As a result, it ensures greater consistency between the outputs of support and local trees compared to approaches that construct entirely new models based solely on incomplete datasets.

In summary, this study utilized VEEM biomarkers with or without missing modalities to train a scandent decision tree for MCI detection. By adopting a feature-based machine learning model, this study aimed to address the black-box limitations commonly associated with deep learning approaches [34], offering clear and interpretable rationales for the model's predictions. The primary objective of this study was to evaluate the scandent decision tree, trained on VEEM biomarkers encompassing VR, EEG, and MRI modalities, in accurately detecting MCI even when one modality was treated as missing data. Finally, this study discussed the practical implications of the trained model in real-world clinical settings.

3 RESEARCH APPROACH AND METHODS

3.1 Participants and Dataset Acquisition

This study recruited a total of 59 participants (average age: 71.49 ± 9.35) including 22 healthy controls (HC) and 37 patients with MCI, from two Hanyang University Hospitals in Guri and Seoul, Republic of Korea, from January 2022 to September 2023. Participants voluntarily enrolled in the study through the hospital community. The inclusion criteria required participants to be able to speak and read Korean and to have no hearing impairments. Participants with a history of brain surgery were excluded. All participants were diagnosed as HC or MCI based on their neuropsychological test results using the Seoul Neuropsychological Screening Battery-Core (SNSB-C) [17]. The diagnosis of MCI was confirmed in accordance with Albert et al.'s criteria [3] when a consensus was reached by two neuropsychologists with 18 and 22 years of experience, respectively. After a full explanation, written consent was obtained from all participants. The experiments in this study were conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of Hanyang University Hospital, Republic of Korea (HYUH-2021-08-020-004).

3.2 VR Biomarkers

The VR biomarkers were collected through the virtual kiosk test, which was developed in our previous study [24] in collaboration with neuropsychologists. The virtual kiosk test is an instrumental

activity of daily living-based task that requires participants to place an order for a combo meal using a kiosk in a fully immersive virtual restaurant. Participants wear a head-mounted VR headset and interact with the virtual kiosk using a hand controller. The virtual kiosk test consists of six sequential action steps: (1) choose a place to eat, (2) choose a burger item, (3) choose a side item, (4) choose a drink item, (5) choose a payment method, and (6) enter a 4-digit payment password. To perform each step, participants have to memorize all verbally delivered instructions before initiating the virtual kiosk test: "The place to eat is a restaurant. Please order a shrimp burger, cheese sticks, and a Coca-Cola using the kiosk. Please use a credit card as the payment method. The card payment password is 6289." During the virtual kiosk test, the hand movement feature, the eye movement feature, and performance features were collected in real time as VR data. Specifically, the hand movement feature include *hand movement speed*; the eye movement feature comprise *proportion of fixation duration*; performance features encompass *time to completion* and *the number of errors* (descriptions of each feature are elucidated in Table A1 in the Appendix).

3.3 EEG Biomarkers

The EEG biomarkers were collected using the steady-state visually evoked potential (SSVEP) method for two minutes while participants were lying in a dimly lit room with their eyes closed. SSVEP was elicited by intermittent photic stimulation, and the corresponding brain activity was recorded. EEG signals were captured using the Comet-Plus XL Lab EEG system (Natus Neurology, Middleton, WI, USA) at a sampling rate of 200 Hz. Visual stimuli were presented at frequencies of 3, 5, 10, 12, 15, and 20 Hz, each consisting of a 10-second photic stimulation followed by a 10-second rest interval. Electrode placement followed the international 10-20 system, including Fp1, Fp2, F3, F4, F7, F8, C3, C4, T3, T4, T5, T6, P3, P4, O1, O2, Fz, Cz, and Pz.

Following the pre-processing steps outlined in our previous study [22], five EEG features were derived and categorized as follows: 1) Lobe Power Ratio (LPR): This category includes two features (i.e., 12PS-POR-1H and 15PS-POR-1H), representing the ratio of power between the parietal and occipital lobes; 2) Lobe Connectivity Ratio (LCR): This category includes two features (i.e., 3PS-POR- α and 5PS-POR- γ), representing connectivity ratios within the parietal and occipital lobes; and 3) Band Connectivity Ratio (BCR): This category includes one feature (i.e., 15PS-P-TBR), capturing connectivity within the parietal lobe based on predefined frequency band ratios.

Each feature was assigned an abbreviation for clarity and consistency. Specifically, "PS" denotes the frequency of photic stimulation, "POR" denotes the ratio of power or connectivity in the parietal and occipital lobes, "H" indicates the specific harmonic in response to visual stimuli, and " α " and " γ " represent connectivity in the 8–12 Hz and 30–50 Hz frequency bands, respectively. "P" represents the set of electrodes within the parietal lobe, while "TBR" refers to the θ/β ratio of band connectivity. For example, "12PS-POR-1H" indicates the power ratio of the fundamental harmonic in the parietal and occipital lobes during 12 Hz photic stimulation. Detailed descriptions of each feature are provided in Table A1 in the Appendix.

3.4 MRI Biomarkers

The MRI biomarkers were acquired using the 3D T1-weighted magnetization-prepared rapid gradient echo technique by utilizing a Philips Ingenia CX 3T scanner. MRI scans were performed at Hanyang University Hospitals in Guri and Seoul following specific protocols. In Guri, the acquisition parameters were echo time/repetition time = 2.9 ms/6.3 ms, flip angle = 9°, field of view = 256 × 256 mm, with 211 slices and a voxel size of 1 × 1 × 1 mm³. In Seoul, the parameters were echo time/repetition time = 4.1 ms/6.9 ms, flip angle = 8°, field of view = 300 × 299 mm, with 170 slices and a voxel size of 0.8 × 0.8 × 1 mm³.

To quantify the volume of the region of interest in the brain, we used AQUA 3.0 software (Neurophet Inc, Seoul, Republic of Korea), and the entire pre-processing procedure was described in our previous study [30]. To account for individual variations in brain size, MRI features were normalized to intracranial volume (ICV), which encompasses total white matter, gray matter, and cerebrospinal fluid volumes [29]. In this study, six MRI features were utilized, including the lateral volumes of the amygdala, hippocampus, and entorhinal cortex, which have been reported as effective features for detecting patients with MCI. Detailed descriptions of each feature are provided in Table A1 in the Appendix.

3.5 Experimental Procedure

Participants completed four assessments as follows: (1) the SNSB-C test for the diagnosis of MCI, (2) a virtual kiosk test designed to collect the VR biomarkers, (3) an SSVEP session to collect the EEG biomarkers, and (4) a T1-weighted MRI session to collect the MRI biomarkers. To mitigate potential order effects, the order of assessments was randomized. The virtual kiosk test was conducted after two practice sessions, allowing participants to familiarize themselves with the virtual environment and test procedures. The SSVEP and T1-weighted MRI sessions were administered by experienced professionals: an EEG technician and a radiologist with 16 years of experience, respectively. Participants were allowed to pause the procedure at any time for any reason, including discomfort or dizziness. The entire experimental process took approximately 90 minutes on average.

3.6 Analyses

We conducted statistical analyses using IBM SPSS Statistics 27 (IBM, Armonk, New York, USA). First, the demographic characteristics (i.e., gender, age, and education level) were compared between HC and patients with MCI using a chi-square test and independent-sample t tests. To identify significant differences in VEEM biomarkers between HC and patients with MCI, independent-sample t tests were conducted. As a result, there were no significant differences in gender, age, and education level between the two groups as shown in Table 1.

3.7 A Multimodal Approach

We implemented a feature-based machine learning approach to ensure both the explainability and reliability of the model's output. The scandent decision tree, a modified version of the standard decision tree, requires training the support tree on a complete

dataset without missing data. Local trees are then trained to replicate the splits of the support tree, using only the modality-specific features that vary between parent and child nodes to counteract the missing data in multimodal biomarkers. In this study, the support tree was trained on the complete VEEM dataset, and one modality among VR, EEG, and MRI was excluded as missing data to test the performance of the trained scandent decision tree. Using the GridSearch library in Python, the optimal tree configurations were identified. For the support tree, the parameters were {max_depth: 4, min_samples_split: 2, min_samples_leaf: 3, criterion: 'gini', splitter: 'random'}. For the local trees, the parameters were {max_depth: 4, min_samples_split: 3, min_samples_leaf: 3, criterion: 'entropy', splitter: 'random'}. Performance evaluation was conducted using 10-fold cross-validation to prevent overfitting, with 80% of the dataset allocated for training and 20% for testing. The evaluation metrics included accuracy, sensitivity, specificity, precision, and F1-score.

4 RESULTS

4.1 VEEM Biomarkers

As shown in Table 2, all 15 features in VEEM biomarkers exhibited significant differences between HC and patients with MCI. Specifically, in VR biomarkers, hand movement speed ($p = .02$) and proportion of fixation duration ($p < .001$) were significantly higher in HC than in patients with MCI, suggesting compromised visual and motor exploration in MCI. Conversely, time to completion ($p = .007$) and the number of errors ($p < .001$) were significantly higher in patients with MCI, indicating prolonged task duration and increased errors compared to HC. Regarding EEG biomarkers, all EEG features were significantly higher in patients with MCI than in HC ($p < .05$). This could imply that the effectiveness of brain connectivity was impaired in patients with MCI. Similarly, all MRI features were significantly lower in patients with MCI than in HC ($p < .05$), indicating atrophy in brain subregions in patients with MCI. Thus, all 15 features were used for the machine learning model for MCI detection.

4.2 Scandent Decision Tree Using Complete and Incomplete VEEM Biomarkers

Using the complete VEEM biomarkers (i.e., four VR features, five EEG features, and six MRI features), we trained the support tree, which serves as the foundation of the scandent decision tree. As shown in Table 3, the model achieved an accuracy of 91.67%, sensitivity of 85.70%, specificity of 100.0%, precision of 93.06%, and an F1-score of 91.72%.

The local trees derived from the support tree were trained under simulated conditions where one of the VR, EEG, or MRI modalities was missing. When VR or EEG biomarkers were excluded, the machine learning model for MCI detection achieved an accuracy of 66.67%, sensitivity of 85.71%, specificity of 40.0%, precision of 66.67%, and an F1-score of 64.58%. When MRI biomarkers were excluded, the model demonstrated improved performance, achieving an accuracy of 83.33%, sensitivity of 100.0%, specificity of 60.0%, precision of 87.04%, and an F1-score of 82.29%, which indicated improved results compared to the condition where VR or EEG biomarkers were excluded.

Table 1: Comparison of demographic characteristics between the two groups

Characteristics	Healthy Controls (N = 22)	Patients with MCI ^a (N = 37)	<i>p</i> value
Gender (female/male)	9/13	21/16	0.36
Age (years)	68.63 ± 10.18	73.18 ± 8.51	0.07
Education level (years)	12.27 ± 4.53	10.13 ± 5.32	0.12

^a MCI: Mild cognitive impairment**Table 2: Comparison of 15 VEEM biomarkers between the two groups**

Biomarkers	Healthy Controls (N = 22)	Patients with MCI ^a (N = 37)	<i>p</i> value
VR biomarkers			
Hand movement speed (m/s)	0.23 ± 0.06	0.17 ± 0.05	.02
Proportion of fixation duration (%)	55.69 ± 13.11	45.72 ± 17.23	< .001
Time to completion (seconds)	39.83 ± 18.74	99.83 ± 84.07	.007
The number of errors	1.77 ± 1.57	3.59 ± 2.80	< .001
EEG biomarkers			
12PS-POR-1H	0.73 ± 0.07	0.80 ± 0.08	< .001
15PS-POR-1H	0.73 ± 0.06	0.79 ± 0.07	.002
3PS-POR- α	0.96 ± 0.04	1.00 ± 0.06	.01
5PS-POR- γ	0.98 ± 0.02	1.00 ± 0.03	.01
15PS-P-TBR	0.92 ± 0.11	0.99 ± 0.11	.02
MRI biomarkers			
Left amygdala (%)	0.15 ± 0.01	0.13 ± 0.02	< .001
Right amygdala (%)	0.24 ± 0.02	0.21 ± 0.03	.005
Left hippocampus (%)	0.23 ± 0.02	0.21 ± 0.03	.002
Right hippocampus (%)	0.11 ± 0.01	0.10 ± 0.01	.003
Left entorhinal cortex (%)	0.12 ± 0.01	0.1 ± 0.01	.002
Right entorhinal cortex (%)	0.12 ± 0.01	0.11 ± 0.01	< .001

^a MCI: Mild cognitive impairment**Table 3: Performance of scandent decision tree using complete and incomplete VEEM biomarkers**

Biomarkers	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-score (%)
VEEM biomarkers	91.67	85.70	100.0	93.06	91.72
VR missing (EEG+MRI)	66.67	85.71	40.0	66.67	64.58
EEG missing (VR+MRI)	66.67	85.71	40.0	66.67	64.58
MRI missing (VR+EEG)	83.33	100.0	60.0	87.04	82.29

5 DISCUSSION

By leveraging VEEM biomarkers and datasets with missing modalities, we trained a scandent decision tree, which achieved the highest performance when all VEEM biomarkers were included, attaining an accuracy of 91.67%, sensitivity of 85.70%, specificity of 100.0%, precision of 93.06%, and F1-score of 91.72%. The second-best performance was observed when the MRI biomarkers were missing (i.e., when the model was trained with VR and EEG biomarkers), whereas the lowest performance was observed when either VR or EEG biomarkers were absent. These findings highlight the efficacy of the scandent decision tree in maintaining stable performance

even with missing data. Considering the variability in clinical environments and the patient burden associated with comprehensive data collection, the proposed multimodal approach employing scandent decision trees offers a significant advantage in handling missing data scenarios.

An interesting finding is that the loss of performance was greater when VR or EEG biomarkers were missing compared to when MRI biomarkers were missing. This finding aligns with previous research, which indicates that behavioral and neurological alterations typically precede structural brain atrophy [30, 31], suggesting that VR and EEG biomarkers may be more effective in detecting early-stage MCI. Additionally, an intriguing result was that the sensitivity was higher in the model using VR and EEG biomarkers than in

the model using complete VEEM biomarkers. This could imply that modalities that are captured during dynamic interactions, such as ordering a combo meal or being exposed to photic stimulation may be more sensitive to changes in early-stage MCI than static structural imaging, such as MRI scans. These insights pave the way for a sequential clinical approach: early-stage MCI could be monitored using VR or EEG biomarkers—which are relatively less invasive—based on factors such as age, subjective complaints, or SNSB-C results. In contrast, mid- to late-stage MCI could be more comprehensively evaluated with complete VEEM biomarkers, including MRI biomarkers, to enable precise diagnosis.

Despite the practical findings, this study has several limitations. First, the small sample size may have introduced variability in performance. Recruiting more participants could contribute to developing more robust models for MCI detection. Second, the study primarily addressed missing data at the modality level. Future work could explore handling feature-level missing data in multimodal models, which may further enhance its applicability. Last, while the decision tree model offers improved interpretability, relying on multiple modalities may increase the cognitive load on clinicians. Future studies could integrate advanced methods, such as large language models, to enhance interpretability and mitigate this burden. Addressing these limitations would improve the model's robustness and clinical relevance, amplifying its utility in diverse clinical settings.

6 CONCLUSION

This study demonstrated that the multimodal approach integrating VEEM biomarkers achieved the highest performance in MCI detection, with an accuracy of 91.67%, sensitivity of 85.70%, specificity of 100.0%, precision of 93.06%, and an F1-score of 91.72%. Notably, the model maintained consistent performance even when specific modalities were missing. This approach holds significant promise for practical application in clinical settings. For early MCI detection, VR and EEG biomarkers can be effectively utilized, while MRI biomarkers might be employed as symptoms advance. Ultimately, these findings have the potential to reduce the financial burden on society and minimize patient discomfort during MCI detection.

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